Earners AI and ML by Aditya Jain sir - Lecture 24

Summary Notes

DPP Solutions

```
numbers = [10, 15, 20]

result = list(map(lambda x: x // 5, numbers))

print(result)
```

```
[2, 3, 4]
```

```
numbers = [3, 6, 9, 12]

result = list(map(lambda x: x + 1, filter(lambda x: x % 6 == 0, numbers)))

print(result)
```

```
[7, 13]
```

```
numbers = [3, 6, 9, 12]


ans = list(filter(lambda x: x % 6 == 0, numbers))

print(ans)
```

```
[6, 12]
```

```
result = list(map(lambda x: x + 1, ans))
```

```
print(result)
```

```
[7, 13]
```

```
marks = [35, 45, 80, 25, 90]

passed = filter(lambda x: x >= 40, marks)

bonus = list(map(lambda x: x + 5, passed))

print(bonus)
```

```
[50, 85, 95]
```

```
marks = [35, 45, 80, 25, 90]

bonus = list(map(lambda x: x + 5, filter(lambda x: x >= 40, marks)))

print(bonus)
```

```
[50, 85, 95]
```

```
numbers = [2, 3, 4]

result = list( map(
```

```
        lambda x: x ** 2,
        filter(lambda x: x > 2, numbers)
    )
)

print(result)
```

```
numbers = [2, 3, 4]

ans1 = list(filter(lambda x: x > 2, numbers))
ans1
```

```
[3, 4]
```

```
result = list( map( lambda x: x ** 2, ans1))
print(result)
```

```
[9, 16]
```

```
numbers = [1, 2, 3, 4, 5]

result= list(map(
        lambda x: x * 10,
        filter(lambda x: x % 2 != 0, numbers)
    )
)

print(result)
```

```
numbers = [1, 2, 3, 4, 5]

ans1 = list(filter(lambda x: x % 2 != 0, numbers))
print(ans1)
```

```
[1, 3, 5]
```

```
result= list(map(lambda x: x * 10,ans1))
print(result)
```

```
[10, 30, 50]
```

File Handling using Python

```
name = input()
print(name)
```

```
AJ
AJ
```

```
# most commonly used types of files in AIML --> text, csv

# csv --> comma separated values.

# Name,Marks,City
# Rahul,95,Nagpur
# Amit,88,Pune
# Neha,90,Delhi
```

```
#syntax
# file = open(filename, mode)
```

```
file = open("sample.txt", "w")
file.write("AJ\n")
file.write("Rahul\n")
file.write("Raj")
file.close()
```

```
file = open("sample.txt", "r")
data = file.read()
print(data)
file.close()
```

```
AJ
Rahul
Raj
```

```
file = open("sample.txt", "r")
data = file.readline()
print(data)
file.close()
```

```
AJ
```

```
file = open("sample.txt", "r")
data = file.readline()
print(data)
data = file.readline()
print(data)
file.close()
```

```
AJ

Rahul
```

```
file = open("sample.txt", "r")
data = file.readlines()
print(data)
file.close()
```

```
['AJ\n', 'Rahul\n', 'Raj']
```

```
for line in data:
  print(line)
```

```
AJ

Rahul

Raj
```

```
file = open("sample.txt", "r")

for line in file:
  print(line)

file.close()
```

```
AJ

Rahul
```

```
Raj
```

```
file = open("sample.txt", "r")

for line in file:
  print(line, end = "")

file.close()
```

```
AJ
Rahul
Raj
```

```
# r -> read
# w -> write
# a -> append
```

```
file = open("sample.txt", "a")
file.write("\nAditya")
file.close()
```

```
file = open("sample.txt", "r")

for line in file:
  print(line)

file.close()
```

```
AJ

Rahul

Raj

Aditya
```

```
file = open("abc.txt", "x")
file.close()
```

```
with open("sample.txt", "r") as file:
  print(file.read())

#this is a more standard way of file handling and there is no need for closing the file.
```

```
AJ
Rahul
Raj
Aditya
```

```
with open("demo.txt", "w") as file:
  file.write("Aditya")
```

```
with open("demo.txt", "a") as file:
  file.write("Jain")
```

```
with open("demo.txt", "r") as file:
  data = file.read()
  print(data)
```

```
AdityaJain
```

CSV

```
import csv

with open("student.csv", "w", newline="") as file:
  writer = csv.writer(file)
  writer.writerow(["Name", "Marks", "City"])
  writer.writerow(["Raj",  90, "Mumbai"])
  writer.writerow(["Rahul", 95, "Hyderabad"])
```

```
#Write a code to read the content from the csv file and display it

with open("student.csv", "r") as file:
  reader = csv.reader(file)

  for row in reader:
    print(row)
```

```
['Name', 'Marks', 'City']
['Raj', '90', 'Mumbai']
['Rahul', '95', 'Hyderabad']
```

```
#take a new record as input from the user and add it to the csv file.
#HW
name = input()
marks = int(input())
city = input()
```